WEATHER REPORT DASHBOARD

Power BI Project Documentation

Built with WeatherAPI.com | Power BI Desktop

Project Name	Weather Report Dashboard
Tool Used	Microsoft Power BI Desktop
Data Source	WeatherAPI.com (REST API)
Cities Covered	Mumbai, Chennai, Hyderabad, Noida, Kolkata, Jaipur
Data Refresh	Manual (Scheduled after publishing)
Documentation	05 July 2026

This document provides a complete technical and functional record of the Weather Report Dashboard built in Power BI Desktop. It covers the project overview, data architecture, data model, DAX measures, visual components, and key design decisions made during development.

1. Project Overview

Purpose, scope, and objectives of the dashboard

The Weather Report Dashboard is an interactive Power BI report that displays live and forecast weather data for six major Indian cities. It is designed to give users a comprehensive at-a-glance view of current conditions, a 7-day forecast, air quality, temperature trends, rainfall, and key daily highlights — all in a single page.

1.1 Objectives

- Display real-time weather data fetched directly from WeatherAPI.com
- Allow users to switch between 6 cities using an interactive slicer
- Show a 7-day forecast with weather icons, max and min temperatures
- Visualise temperature trends and rainfall via charts
- Display Air Quality Index (AQI) with color-coded category and health suggestion
- Show sunrise and sunset times with a visual arc illustration
- Present key highlights — Today's High, Low, Chance of Rain, UV Index
- Keep the design consistent with a professional teal-and-navy color theme

1.2 Cities Covered

City	State	API Parameter
Mumbai	Maharashtra	q=Mumbai
Chennai	Tamil Nadu	q=Chennai
Hyderabad	Telangana	q=Hyderabad
Noida	Uttar Pradesh	q=Noida
Kolkata	West Bengal	q=Kolkata
Jaipur	Rajasthan	q=Jaipur

2. Data Source

WeatherAPI.com REST API integration

All weather data is sourced from **WeatherAPI.com**, a commercial weather data provider offering real-time, forecast, and historical weather data via a REST API. The free tier provides up to 1,000,000 API calls per month, which is sufficient for scheduled hourly refreshes across 6 cities.

2.1 API Endpoint

```
https://api.weatherapi.com/v1/forecast.json
  ?key={API_KEY}
  &q;={CITY_NAME}
  &days;=7
  &aqi;=yes
  &alerts;=no
```

2.2 API Parameters

Parameter	Value	Purpose
key	{API_KEY}	Authentication — unique API key from WeatherAPI account
q	City name	Location query — one per query (6 separate API calls)
days	7	Number of forecast days to return
aqi	yes	Include Air Quality Index data in response
alerts	no	Exclude weather alerts (not yet implemented)

2.3 Response Structure

The API returns a nested JSON object with three main sections: **location** (city metadata), **current** (live conditions), and **forecast** (7-day daily and hourly breakdown). Power Query flattens this nested structure into tabular form through a series of expand steps.

2.4 Data Refresh

- Currently: manual refresh via Home → Refresh in Power BI Desktop
- After publishing to Power BI Service: scheduled refresh can be configured
- Recommended refresh frequency: every 1 hour
- Each refresh makes 6 API calls (one per city)
- Last Updated timestamp updates automatically from current.last_updated field

2.5 Security Note

Important: The API key is currently embedded in each of the 6 Power Query queries as a hardcoded string. Best practice is to move it into a Power Query Parameter (Home → Manage Parameters) so it can be updated in one place and is not directly visible in query source code.

3. Data Model

Tables, columns, and relationships

The data model consists of four active tables used by visuals, plus several staging tables used only during the Power Query transformation process. The staging tables should have Load disabled to reduce file size and keep the Data pane clean.

3.1 Active Tables

Table	Source	Key Fields	Used For
Locations	Calculated (DAX)	location.name, location.region, Display Name	City slicer, relationships hub
current	Master_table	temp_c, feelslike_c, humidity, wind_kph, condition, AQI fields	Hero card, AQI, Key Highlights
forecast_day	Master_table	date, maxtemp_c, mintemp_c, totalprecip_mm, chance_of_rain, astro.*	7-Day Forecast, Trend chart, Rainfall, Sunrise/Sunset
forecast_hour	Master_table	time, temp_c, condition, wind_kph	Future hourly breakdown (available for use)

3.2 Staging Tables (Load should be disabled)

Table	Purpose
weather_report_1–6	Raw API response per city — expanded and flattened by Power Query
Master_table	Combines all 6 weather_report tables via Table.Combine

3.3 Relationships

From	To	Key Column	Type
Locations[location.name]	current[location.name]	location.name	1 → Many (Active)
Locations[location.name]	forecast_day[location.name]	location.name	1 → Many (Active)
Locations[location.name]	forecast_hour[location.name]	location.name	1 → Many (Active)

3.4 Calculated Columns

Table	Column Name	Formula	Purpose
Locations	Display Name	location.name & ", " & location.region	Combined city+state for slicer display
current	Icon URL	"https:" & current.condition.icon	Full URL for live weather icon (if not already https)
forecast_day	Day Short	FORMAT(date, "ddd")	Short day name (Mon/Tue) for chart X-axis

4. DAX Measures

All measures created for the dashboard

Measure Name	Table	Formula (simplified)	Purpose
curr_temp_c	_measures	MAX(current[current.temp_c])	Current temperature used in hero card — displayed with °C format string applied via Measure tools
Last_Updated_C	_measures	FORMAT(MAX(current[current.last_updated]), "dd MMM yyyy | hh:mm AM/PM")	Formatted last updated timestamp for header badge
Max Temp	_measures	MAX(forecast_day[forecast.forecastday.day.maxtemp_c])	Daily maximum temperature for trend line chart
Min Temp	_measures	MIN(forecast_day[forecast.forecastday.day.mintemp_c])	Daily minimum temperature for trend line chart
Today High	_measures	CALCULATE(MAX(forecast_day[...maxtemp_c]), FIRSTDATE(forecast_day[...date]))	Today's forecast high temperature for Key Highlights
Today Low	_measures	CALCULATE(MIN(forecast_day[...mintemp_c]), FIRSTDATE(forecast_day[...date]))	Today's forecast low temperature for Key Highlights
Chance of Rain	_measures	CALCULATE(MAX(forecast_day[...daily_chance_of_rain]), FIRSTDATE(forecast_day[...date]))	Today's chance of rain percentage for Key Highlights
UV Index Display	_measures	FORMAT(SELECTEDVALUE(current[current.uv]),"0") & " (" & SWITCH(TRUE(),...) & ")"	UV value with category label e.g. '6 (High)'
Today Sunrise	_measures	CALCULATE(SELECTEDVALUE(forecast_day[...astro.sunrise]), FIRSTDATE(forecast_day[...date]))	Today's sunrise time text for Sunrise & Sunset card
Today Sunset	_measures	CALCULATE(SELECTEDVALUE(forecast_day[...astro.sunset]), FIRSTDATE(forecast_day[...date]))	Today's sunset time text for Sunrise & Sunset card
AQI Value	_measures	SWITCH(SELECTEDVALUE(current[...us-epa-index]), 1,25, 2,75, 3,125, 4,175, 5,250, 6,400)	Representative AQI number (25/75/125...) from EPA category code
AQI_Status	_measures	SWITCH(SELECTEDVALUE(current[...us-epa-index]), 1,'Good', 2,'Moderate', ...)	AQI category text label for display
AQI Color	_measures	SWITCH(SELECTEDVALUE(current[...us-epa-index]), 1,'#7EBC45', 2,'#E7CB23', ...)	Hex color string for conditional formatting of AQI gauge visual
AQI Suggestion	_measures	SWITCH(SELECTEDVALUE(current[...us-epa-index]), 1,'Air is clean and healthy', ...)	Health advice text based on AQI category
Wind_Speed	_measures	SELECTEDVALUE(current[current.wind_kph])	Current wind speed for hero card bottom strip

Wind_Speed	_measures	SELECTEDVALUE(current[current.wind_kph])	Current wind speed for hero card bottom strip

4.1 Key DAX Patterns Used

Pattern	Function	Why Used
Single-value selection	SELECTEDVALUE()	Returns one value based on slicer — fails gracefully if multiple rows
Date anchoring	FIRSTDATE()	Gets today's row from forecast_day without relying on TODAY() date match
Conditional branching	SWITCH(TRUE(), ...)	Cleaner than nested IF for multiple conditions
Context override	CALCULATE()	Changes filter context to specific date or city
Text formatting	FORMAT()	Converts numbers/dates to display strings

5. Dashboard Components

Visual-by-visual breakdown

Section	Visual Types	Content / Fields
Header	Text boxes, Image, Slicer, Card	Title (WEATHER REPORT), leaf branch decoration, city slicer (Display Name), Last Updated badge
Current Weather Hero Card	Rectangle shape, Card visuals, Image	City name, live temperature (curr_temp_c), condition text, weather icon (dynamic URL), Feels Like, Humidity, Wind Speed — all on teal #2DA5A8 background
7-Day Forecast	Rectangle shapes, Image visuals, Card visuals, Text boxes	7 columns each showing: day name, weather icon (condition.icon URL), max and min temperature for that forecast day
Temperature Trend Chart	Line chart	X-axis: Day Short column (Mon–Sun), Y-axis: Max Temp (deep teal #217081) and Min Temp (light teal #9FD9DC) lines with markers and data labels
Air Quality Index	Native Gauge, Card visuals, Text boxes	Gauge filled with AQI Color measure, AQI Value number, AQI_Status label, AQI Suggestion health text, 6 pollutant Card visuals (PM2.5, PM10, SO2, NO2, CO, O3)
Sunrise & Sunset	Image, Card visuals, Text boxes	arc_dotted.png decorative arc, icon_sunrise.png + Today Sunrise Card, icon_sunset.png + Today Sunset Card
Rainfall Chart	Clustered column chart	X-axis: Day Short, Y-axis: totalprecip_mm, bars colored teal #2DA5A8, data labels on
Key Highlights	Image visuals, Text boxes, Card visuals, Shape lines	4 rows: Today's High (thermometer_high icon), Today's Low (thermometer_low icon), Chance of Rain (rain icon), UV Index Display (sun icon) — values right-aligned, divider lines between rows

6. Color Palette & Design

Visual design tokens and typography

Token	Hex	Used For
Primary Teal	#2DA5A8	Hero card background, bars, badges, slicer
Deep Teal	#217081	Max Temp line, dark accents
Light Teal	#9FD9DC	Min Temp line, subtle highlights
Navy	#0B1830	Headlines, big numbers, body text
Muted Gray	#9AA5B5	Secondary labels, captions
Light Gray	#E5E9EF	Divider lines, card borders
Canvas Background	#DCE7F0	Page background
Sun Yellow	#FEC516	Sun icons, rays
AQI Good	#7EBC45	Green — AQI 1 (Good)
AQI Moderate	#E7CB23	Yellow — AQI 2 (Moderate)
AQI USG	#FDAB4F	Orange — AQI 3 (Unhealthy for Sensitive)
AQI Unhealthy	#EE6D5E	Red — AQI 4 (Unhealthy)
AQI Very Unhealthy	#B23A48	Dark Red — AQI 5 (Very Unhealthy)
AQI Hazardous	#7A1F2B	Very Dark Red — AQI 6 (Hazardous)

6.1 Typography

Headlines / Title	Poppins Bold — 28–32pt — Navy #0B1830
Big numbers (temp)	Poppins Bold — 48pt — White (on teal card)
Subheadings	Poppins SemiBold — 14–16pt — Navy
Body / Labels	Segoe UI / Inter Regular — 10–13pt — Gray
Data values	Poppins Bold — 16–20pt — Navy, right-aligned

6.2 Card Design Rules

- All cards use white fill (#FFFFFF) with rounded corners ~16px
- Soft drop shadow on every card — ~15% opacity, small blur
- 16–20px consistent gutters between all cards
- No visible borders (border off in Format visual)
- Background off on all visuals inside cards so rectangle color shows through

7. AQI Reference

US EPA AQI category system used in this dashboard

WeatherAPI returns the Air Quality Index as a 1–6 category code based on the US EPA standard. This dashboard translates that code into a display number, status label, color, and health suggestion using DAX SWITCH measures.

Code	Category	Display Number	Color	Health Suggestion
1	Good	25	#7EBC45 (Green)	Air is clean and healthy
2	Moderate	75	#E7CB23 (Yellow)	Acceptable air quality, stay active
3	Unhealthy for Sensitive	125	#FDAB4F (Orange)	Sensitive groups reduce outdoor time
4	Unhealthy	175	#EE6D5E (Red-Orange)	Limit prolonged outdoor exertion
5	Very Unhealthy	250	#B23A48 (Dark Red)	Avoid outdoor activity if possible
6	Hazardous	400	#7A1F2B (Maroon)	Stay indoors, wear mask if outside

Note: The display numbers (25, 75, 125...) are representative midpoints of each EPA category range, not precise calculated values. For a more accurate score, a PM2.5-based AQI formula can be applied to the raw pm2_5 concentration field from the API — this was discussed during development but the category-code approach was chosen for simplicity and reliability.

8. Known Issues & Limitations

Issue	Impact	Resolution / Workaround
API key hardcoded in 6 queries	Security risk if file is shared	Move to a Power Query Parameter — Home → Manage Parameters
Staging tables still loaded	Larger file size, cluttered Data pane	Right-click weather_report_1–6 and Master_table → Disable Load
sunrise/sunset stored as text, not time type	Cannot do time arithmetic on these fields	Acceptable — display-only use case, text format is fine
AQI gauge is native Gauge visual (single color)	Cannot show multi-color band arc natively	gauge_arc.png image used as overlay — or install Deneb for full arc
Privacy level required set to Ignore	Minor security setting change	Set all queries to Public explicitly in query Properties instead
Weather Alerts not implemented	Alert panel shows no live data	Add &alerts;=yes to API URLs and expand the alerts.alert list in Power Query
No sidebar navigation built	Single-page, no nav panel	Add rectangle + Buttons + mountain_river.png as future enhancement

9. Future Enhancements

• Weather Alerts Panel
Add &alerts;=yes to API calls, expand the alerts.alert nested list in Power Query, bind to a Table visual showing headline, severity, and effective date

• Sidebar Navigation
Add a vertical navigation panel with logo, menu buttons (Overview/Forecast/Analytics), and the mountain_river.png landscape illustration

• Multi-color AQI Gauge
Install Deneb visual from AppSource and use the Vega-Lite semicircle spec for a proper four-color banded gauge with a live needle

• Hourly Forecast Page
Add a second report page using forecast_hour table to show temperature, rain chance, and wind by hour for the selected day

• Historical Comparison
Use WeatherAPI's History endpoint to pull last 7 days of data and compare actual vs forecast accuracy

• Mobile Layout
Create a Phone layout version of the report in Power BI Desktop (View → Mobile layout) for Power BI mobile app users

• Scheduled Refresh
Publish to Power BI Service and configure hourly scheduled refresh so the dashboard updates automatically without manual intervention

• API Key Security
Move API key to a Power Query Parameter and document the rotation process so the key can be updated without opening the PBIX file

10. Lessons Learned

Key DAX and Power BI concepts mastered during this project

Concept	What Was Learned
SELECTEDVALUE vs MAX	MAX ignores slicer context and returns the global maximum. SELECTEDVALUE respects the slicer and returns the value for the selected city only. Any measure that should change when the city slicer changes must use SELECTEDVALUE.
Measure vs Column	Columns store a value per row and are computed once on load. Measures are formulas evaluated fresh in the context of each visual. Single-value display visuals (Card) need measures; multi-row visuals (charts) can use columns.
FIRSTDATE for today anchoring	TODAY() in DAX does not match date strings stored as text in the API data. FIRSTDATE(forecast_day[date]) always returns the earliest date in the table — which is always today since WeatherAPI's forecast starts from the current day.
Power Query data types	WeatherAPI returns sunrise/sunset as text like '06:05 AM'. Power Query's auto-detect incorrectly assigns type time, which breaks on 'No moon' values. Always verify auto-assigned types and override to type text for string time fields.
Image URL data category	A text column containing an image URL only renders as a picture in Power BI if its Data Category is set to Image URL in Column tools. Without this, it shows as plain text in Table visuals.
Conditional formatting via fx	The fx button next to color pickers in Format visual allows binding colors to a measure — this is how AQI gauge color changes automatically per city.
Privacy levels	Combining multiple queries (Table.Combine in Master_table) requires either setting each query's privacy level to Public, or setting the file to Ignore privacy levels. This is a configuration issue, not a data or code error.

11. References & Resources

Resource	URL / Location
WeatherAPI Documentation	https://www.weatherapi.com/docs/
Power BI Desktop Download	https://powerbi.microsoft.com/desktop
DAX Reference Guide	https://dax.guide
Deneb Custom Visual	https://deneb-viz.github.io
Vega-Lite Documentation	https://vega.github.io/vega-lite
Poppins Font (Google Fonts)	https://fonts.google.com/specimen/Poppins
Inter Font (Google Fonts)	https://fonts.google.com/specimen/Inter
US EPA AQI Breakpoints	https://www.airnow.gov/aqi/aqi-basics

Document End

This documentation was prepared as part of the Weather Report Dashboard project. All data is sourced from WeatherAPI.com and refreshed on demand. The dashboard was built entirely in Microsoft Power BI Desktop using Power Query M, DAX, and native/custom visuals.